DIGITAL ELECTRONICS

Practice Assignment - Part-wise

Reference level: Problems are designed around the standard Digital Design topics covered in M. Morris Mano's *Digital Design*. They are original practice questions, not reproduced textbook questions.

Student Name	_____
Roll No.	_____
Date	_____
Instructor	_____

Instructions

• Show important steps, Boolean simplifications, truth tables, K-maps, and timing diagrams where applicable.

• Clearly label inputs, outputs, states, select lines, and don't-care conditions.

• For design questions, give the final logic/circuit and a short explanation.

• Questions progress from basic practice to design-oriented problems.

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PART I - NUMBER SYSTEMS & DIGITAL CODES

1. Convert 45, 173, and 625 from decimal to binary, octal, and hexadecimal.

2. Convert 101101.101 (binary) to decimal and hexadecimal.

3. Convert 3A7 (hexadecimal) to binary and decimal.

4. Perform 10110110 + 01101101 using binary addition and show all carries.

5. Perform 11001010 - 01011101 using 2's-complement subtraction.

6. Represent -37 in 8-bit signed-magnitude, 1's-complement, and 2's-complement forms.

7. Determine the range of an n-bit unsigned number and an n-bit 2's-complement number. Apply for n = 8 and n = 12.

8. Perform 3F2(hex) + A7C(hex) and 800(hex) - 2D5(hex).

9. Convert 0.6875(decimal) to binary and 19.625(decimal) to binary.

10. Explain signed overflow. Determine whether overflow occurs for 01110110 + 01011101 and for 10010110 + 10100101.

11. Convert decimal 59 to BCD, Excess-3, and Gray code.

12. Convert Gray code 1101011 to binary, then convert the result back to Gray code.

13. Design a 4-bit binary-to-Gray converter and derive the Boolean equation for each Gray output.

14. Explain why 1010 through 1111 are invalid BCD codes and determine how many invalid 4-bit combinations exist.

15. A 6-bit counter is used in a digital system. Find its number of states and maximum unsigned value. Repeat for signed 2's complement.

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PART II - BOOLEAN ALGEBRA & COMBINATIONAL CIRCUITS

16. Simplify F = A'B + AB + AB' using Boolean algebra.

17. Minimize F(A,B,C) = Sigma m(1,2,3,5,7) using a 3-variable K-map.

18. Minimize F(A,B,C,D) = Sigma m(0,2,3,6,7,8,10,11,14,15) using a 4-variable K-map.

19. For F = AB + A'C + BC, use the consensus theorem to determine whether BC is redundant.

20. Implement F = (A+B)(C+D) using only NAND gates.

21. Implement XOR using only 2-input NAND gates. Find the minimum gate count for your design.

22. Construct the truth table and equations for a half adder.

23. Construct a full adder using two half adders and an OR gate. Derive Sum and Carry.

24. Design a 4-bit ripple-carry adder using full adders. Explain carry propagation delay.

25. Design a 4-to-1 multiplexer using basic gates. Write its output equation.

26. Use a 4-to-1 multiplexer to implement F(A,B,C) = Sigma m(1,2,5,7), using A and B as select inputs.

27. Design a 3-to-8 decoder with an enable input. Write all output equations.

28. Explain why binary-to-BCD conversion generally requires logic rather than direct wiring.

29. Design a 2-bit magnitude comparator and derive A>B, A=B, and A<B equations.

30. A system must select one of eight sensor inputs and route it to one output. Select a suitable combinational circuit and determine the number of select lines.

31. Design a 4-bit parity generator and a 5-bit even-parity checker using XOR/XNOR logic.

32. Explain how don't-care conditions can be used in K-map minimization and provide a small example.

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PART III - SEQUENTIAL CIRCUITS

33. Explain the difference between combinational and sequential circuits. Give two examples of each.

34. Compare SR, JK, D, and T flip-flops in terms of inputs, next-state behavior, and applications.

35. Draw the characteristic and excitation tables for a JK flip-flop. Explain the toggle condition.

36. A D flip-flop has D = Q'. Starting with Q=0, determine Q for the next six active clock edges.

37. Design a synchronous 2-bit up counter using T flip-flops and derive the T-input equations.

38. Design a 3-bit synchronous binary up counter. Show the state sequence and T-input equations.

39. Design a MOD-6 counter. Show valid/unused states and a method for recovery from an unused state.

40. Compare ripple and synchronous counters, focusing on propagation delay and applications.

41. Design a shift register that can shift data both left and right. Identify the control signal and data path.

42. A 4-bit SIPO register starts at 0000. Shift in 1,0,1,1, one bit per clock. Show the register contents after every clock.

43. Design an overlapping sequence detector for 1011. Draw the state diagram and mark the output condition.

44. Convert the sequence detector into a Moore-machine design and explain the difference in output timing.

45. Draw a generic synchronous FSM block diagram and explain present-state register, next-state logic, and output logic.

46. An FSM has states S0,S1,S2. For X=1: S0->S1, S1->S2, S2->S0. For X=0, each state holds. Create the state table and choose a binary state assignment.

47. Explain setup time, hold time, clock-to-Q delay, and metastability.

48. Design a 4-bit register that loads parallel data when LOAD=1 and holds its value when LOAD=0 using a 2-to-1 MUX concept.

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PART IV - MIXED DESIGN & VERIFICATION PRACTICE

49. Design a digital system that accepts two 2-bit numbers and produces their sum. Identify the required blocks and data flow.

50. Design a counter with the sequence 0,1,3,2,6,7,5,4, then repeat. Create the state table and propose a flip-flop implementation.

51. Design a Boolean circuit that activates an alarm when at least three of four input conditions are TRUE.

52. Design a 4-bit one-hot to 2-bit binary encoder. State your assumptions for invalid inputs.

53. Create a 4-bit ALU concept supporting ADD, AND, OR, and XOR. Draw the block-level architecture.

54. Design an overlapping detector for the input pattern 110. Explain why overlap matters.

55. For a 4-bit synchronous counter, write verification checks for reset, every state transition, rollover, and illegal-state recovery.

56. Mini project: Combine a 4-bit counter, decoder, and seven-segment interface to make a 0-9 counter. Identify blocks, interfaces, and major verification cases.

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SUGGESTED PRACTICE ORDER

Stage	Topic	Questions
1	Number systems and codes	1-15
2	Boolean algebra and K-maps	16-20
3	Combinational circuits	21-32
4	Flip-flops and counters	33-42
5	FSM and timing	43-48
6	Mixed design and verification	49-56

Reference note: The assignment follows the common Digital Design progression associated with M. Morris Mano, but the practice questions above are newly written rather than copied from the textbook.

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